

ALGORITMO PRIMA DUALE

$$\begin{aligned} \min \quad & 8x_1 - 5x_2 \\ \text{sv} \quad & x_1 + x_2 \leq 6 \\ & 2x_1 - 3x_2 \geq 4 \\ & x_1 \geq 0, x_2 \leq 0 \end{aligned}$$

forma standard

$$\begin{aligned} \min \quad & 8x_1 + 5x_2 \\ \text{sv} \quad & x_1 - x_2 + s_1 = 6 \\ & 2x_1 + 3x_2 - s_2 = 4 \\ & x_1, x_2, s_1, s_2 \geq 0 \end{aligned}$$

duale

$$\begin{aligned} \max \quad & w(z) = 6y_1 + 4y_2 \\ \text{sv} \quad & y_1 + 2y_2 \leq 8 \\ & -y_1 + 3y_2 \leq 5 \\ & y_1 \leq 0 \\ & -y_2 \leq 0 \\ & y_1, y_2 \in \mathbb{R} \end{aligned}$$

duale in forma standard

$$\begin{aligned} \text{sv} \quad & y_1 + 2y_2 + sd_1 = 8 \\ & -y_1 + 3y_2 + sd_2 = 5 \\ & y_1 + sd_3 = 0 \\ & -y_2 + sd_4 = 0 \end{aligned}$$

punto ammissibile scelgo un punto $y^{(0)}$ ammissibile per il problema duale
 $y^{(0)} = \begin{bmatrix} -5 \\ 0 \end{bmatrix}$

complementarità sostituisco il punto $y^{(0)} = \begin{bmatrix} -5 \\ 0 \end{bmatrix}$ nel problema duale in forma standard

$$\begin{aligned} -5 + 0 + sd_1 &= 8 \rightarrow sd_1 = 13 \\ -5 + 0 + 3sd_2 &= 5 \rightarrow sd_2 = 0 \\ -5 + sd_3 &= 0 \rightarrow sd_3 = 5 \\ 0 + sd_4 &= 0 \rightarrow sd_4 = 0 \end{aligned}$$

FORM. RISTR.
 $x_1 = 0, x_2 = 0$

$$\begin{aligned} \min \quad & s_3 + s_4 \\ \text{sv} \quad & -x_2 + s_3 = 6 \\ & 3x_2 + s_2 + s_4 = 4 \quad \rightarrow x_2 \leq 0, s_2 \geq 0, s_3, s_4 \geq 0 \end{aligned}$$

TABULAU

	x_2	s_2	s_3	s_4	b
-2	0	0	1	1	0
s_3	-1	0	1	0	6
s_4	3	-1	0	1	4

$s_3 = 6$
 $s_4 = 4$
 $z = 10$

DUALE DEL PRIM RISTR

$$\begin{aligned} \max \quad & 6 + 4 \\ \text{sv} \quad & -1u_1 + 3u_2 \leq 0 \geq 0 \\ & -u_1 \leq 0 \\ & u_1 \leq -1 \\ & u_2 \leq -1 \\ & u_1, u_2 \text{ libere} \end{aligned}$$

$$\begin{array}{cc} 1 & 1 \\ \frac{1}{3} & \frac{1}{2} \end{array} \quad \begin{array}{c} 1 \\ 10 \end{array}$$

SOLU2 AMMISS $u = [1]$ $\min = \frac{[8 - [-5 + 0]] \begin{bmatrix} 1 \\ 2 \end{bmatrix}}{[1 \ 1] \begin{bmatrix} 1 \\ 2 \end{bmatrix}}, \frac{[0 - [-5 + 0]] \begin{bmatrix} 1 \\ 0 \end{bmatrix}}{[1 \ 1] \begin{bmatrix} 1 \\ 0 \end{bmatrix}}$

$$\left[\frac{13}{3}, \frac{5}{1} \right] \rightarrow \min = 13, 13$$

$$\begin{aligned} \min \quad & x_1 + 6x_2 - 4x_3 + 5x_4 \\ \text{s.t.} \quad & 5x_1 - 4x_2 + 13x_3 + x_4 = 20 \\ & x_1 - x_2 + 5x_3 + x_4 = 8 \\ & x_i \geq 0 \end{aligned}$$

Duale $\max \quad 20y_1 + 8y_2$

$$\begin{aligned} 5y_1 + y_2 &\leq 1 \\ -4y_1 - y_2 &\leq 6 \\ 13y_1 + 5y_2 &\leq -4 \\ y_1 + y_2 &\leq 5 \\ y_i & \text{ libere} \end{aligned}$$

soluzione ammissibile $\bar{y} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$

complementarità

$$\begin{aligned} -5 - 2 + s_1 &= -1 \rightarrow s_1 = 8 \\ +4 + 2 + s_2 &= 6 \rightarrow s_2 = 0 \\ -13 - 10 + s_3 &= -4 \rightarrow s_3 = -6 \\ -1 - 2 + s_4 &= 5 \rightarrow s_4 = 8 \end{aligned}$$

Formula ristretto del primale

$$\begin{aligned} \min \quad & s_1 + s_2 \\ \text{s.t.} \quad & -4x_2 + s_1 = 20 \\ & -x_2 + s_2 = 8 \\ & x_2, s_1, s_2 \geq 0 \end{aligned}$$

	x_2	s_1	s_2	$\bar{b}$
$-z$	0	1	1	0
s_1	-4	1	0	20
s_2	-1	0	1	8

calcolo dell'ottimo

$$\begin{aligned} z &= 20 \\ s_2 &= 8 \\ z &= 28 \end{aligned}$$

valore del probl. ristretto

$$\begin{aligned} \max \quad & 20u_1 + 8u_2 \\ -4u_1 - u_2 &\leq 0 \\ u_1 &\leq 1 \\ u_2 &\leq 1 \\ u_1, u_2 & \text{ libere} \end{aligned}$$

soluz. ammissibile $u = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} -1 \\ -2 \end{bmatrix} + \begin{bmatrix} 8 \\ 8 \end{bmatrix} = \begin{bmatrix} -1 + \frac{8}{5} \\ -2 + \frac{8}{5} \end{bmatrix} = \begin{bmatrix} -\frac{1}{5} \\ -\frac{10}{5} \end{bmatrix}$$

$$\begin{aligned} & \begin{bmatrix} -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 1 \end{bmatrix} \\ & -5 - 2 = -7 \\ & 1 + 7 = 8 \end{aligned}$$

$$\begin{aligned} & \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} \\ & 5 + 1 = 6 \end{aligned}$$

$$\begin{aligned} & \begin{bmatrix} -5 \\ 0 \end{bmatrix} + \frac{13}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ & \begin{bmatrix} -5 \\ 0 \end{bmatrix} + \begin{bmatrix} 13/3 \\ 13/3 \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} -2 \\ 3 \\ 13 \\ 3 \end{bmatrix}$$

$$\frac{8 - \begin{bmatrix} -5 \\ -10 \end{bmatrix}}{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}$$